Nama : glenn mathias d.s
Nim : A11.2025.16608

Lat1
code index.php

```php
<?php
include "Mahasiswa.php";
include "MataKuliah.php";

$mahasiswa_1 = new Mahasiswa();
$makul_1 = new MataKuliah();
$makul_2 = new MataKuliah();

$makul_1->setData("A11.12345", "Pemrograman Berbasis Web");
$makul_2->setData("A11.12346", "Pemrograman Web Lanjut");

$mahasiswa_1->setData("A11.2023.12345", "Aprilyani Nur Safitri", [$makul_1, $makul_2]);

echo "<pre>";
print_r($mahasiswa_1);
echo "</pre>";
```

Mahasiswa.php

```php
<?php
class Mahasiswa
{
    private $nim, $nama, $matkul;

    public function setData($nim, $nama, $matkul)
    {
        $this->nim = $nim;
        $this->nama = $nama;
        $this->matkul = $matkul;
    }

    public function getData()
    {
        return [
            $this->nim,
            $this->nama,
            $this->matkul,
        ];
    }
}
```

MataKuliah.php

```php
<?php
class MataKuliah
{
    private $kode;
    private $nama;

    public function setData($kode, $nama)
    {
        $this->kode = $kode;
        $this->nama = $nama;
    }

    public function getData()
    {
        return [$this->kode, $this->nama];
    }
}
```

Hasilnya localhost/belajar-oop/lat1

```
Mahasiswa Object
(
    [nim:Mahasiswa:private] => A11.2023.12345
    [nama:Mahasiswa:private] => Aprilyani Nur Safitri
    [matkul:Mahasiswa:private] => Array
        (
            [0] => MataKuliah Object
                (
                    [kode:MataKuliah:private] => A11.12345
                    [nama:MataKuliah:private] => Pemrograman Berbasis Web
                )

            [1] => MataKuliah Object
                (
                    [kode:MataKuliah:private] => A11.12346
                    [nama:MataKuliah:private] => Pemrograman Web Lanjut
                )

        )

)
```

Lat2

code index.php

```php
<?php
include "Mahasiswa.php";
include "MataKuliah.php";

$mahasiswa_1 = new Mahasiswa();
$makul_1 = new MataKuliah();
$makul_2 = new MataKuliah();

$makul_1->setData("A11.12345", "Pemrograman Berbasis Web");
$makul_2->setData("A11.12346", "Pemrograman Web Lanjut");

$mahasiswa_1->setData("A11.2023.12345", "Aprilyani Nur Safitri", [$makul_1, $makul_2]);

echo "<pre>";
print_r($mahasiswa_1);
print_r($mahasiswa_1->nama);
print_r($mahasiswa_1->nim);
echo "</pre>";
```

code Mahasiswa.php

```php
<?php
class Mahasiswa
{
    private $nim, $matkul;
    public $nama;

    public function setData($nim, $nama, $matkul)
    {
        $this->nim = $nim;
        $this->nama = $nama;
        $this->matkul = $matkul;
    }

    public function getData()
    {
        return [
            $this->nim,
            $this->nama,
            $this->matkul,
        ];
    }
}
```

code MataKuliah.php

```php
<?php
class MataKuliah
{
    private $kode;
    private $nama;

    public function setData($kode, $nama)
    {
        $this->kode = $kode;
        $this->nama = $nama;
    }

    public function getData()
    {
        return [$this->kode, $this->nama];
    }
}
```

Hasilnya localhost/belajar-oop/lat2

```
Mahasiswa Object
(
    [nim:Mahasiswa:private] => A11.2023.12345
    [matkul:Mahasiswa:private] => Array
        (
            [0] => MataKuliah Object
                (
                    [kode:MataKuliah:private] => A11.12345
                    [nama:MataKuliah:private] => Pemrograman Berbasis Web
                )

            [1] => MataKuliah Object
                (
                    [kode:MataKuliah:private] => A11.12346
                    [nama:MataKuliah:private] => Pemrograman Web Lanjut
                )

        )

    [nama] => Aprilyani Nur Safitri
)
Aprilyani Nur Safitri

Fatal error:  Uncaught Error: Cannot access private property Mahasiswa::$nim in D:\xampp\htdocs\belajar-oop\lat2\index.php:17
Stack trace:
#0 {main}
  thrown in D:\xampp\htdocs\belajar-oop\lat2\index.php on line 17
```

Lat3

code Mahasiswa.php (public $umur)

```php
<?php
class Mahasiswa
{
    private $nim, $nama, $matkul, $tgl_lahir;
    public $umur;

    public function setData($nim, $nama, $matkul, $tgl_lahir)
    {
        $this->nim = $nim;
        $this->nama = $nama;
        $this->matkul = $matkul;
        $this->tgl_lahir = $tgl_lahir;
        $this->setUmur();
    }

    public function setUmur()
    {
        $this->umur = date("Y") - substr($this->tgl_lahir, 0, 4);
    }

    public function getData()
    {
        return [
            $this->nim,
            $this->nama,
            $this->matkul,
            $this->tgl_lahir,
            $this->umur,
        ];
    }
}
```

code Mahasiswa.php (private $umur)

```php
<?php
class Mahasiswa
{
    private $nim, $nama, $matkul, $tgl_lahir;
    private $umur;

    public function setData($nim, $nama, $matkul, $tgl_lahir)
    {
        $this->nim = $nim;
        $this->nama = $nama;
        $this->matkul = $matkul;
        $this->tgl_lahir = $tgl_lahir;
        $this->setUmur();
    }

    public function setUmur()
    {
        $this->umur = date("Y") - substr($this->tgl_lahir, 0, 4);
    }

    public function getData()
    {
        return [
            $this->nim,
            $this->nama,
            $this->matkul,
            $this->tgl_lahir,
            $this->umur,
        ];
    }
}
```

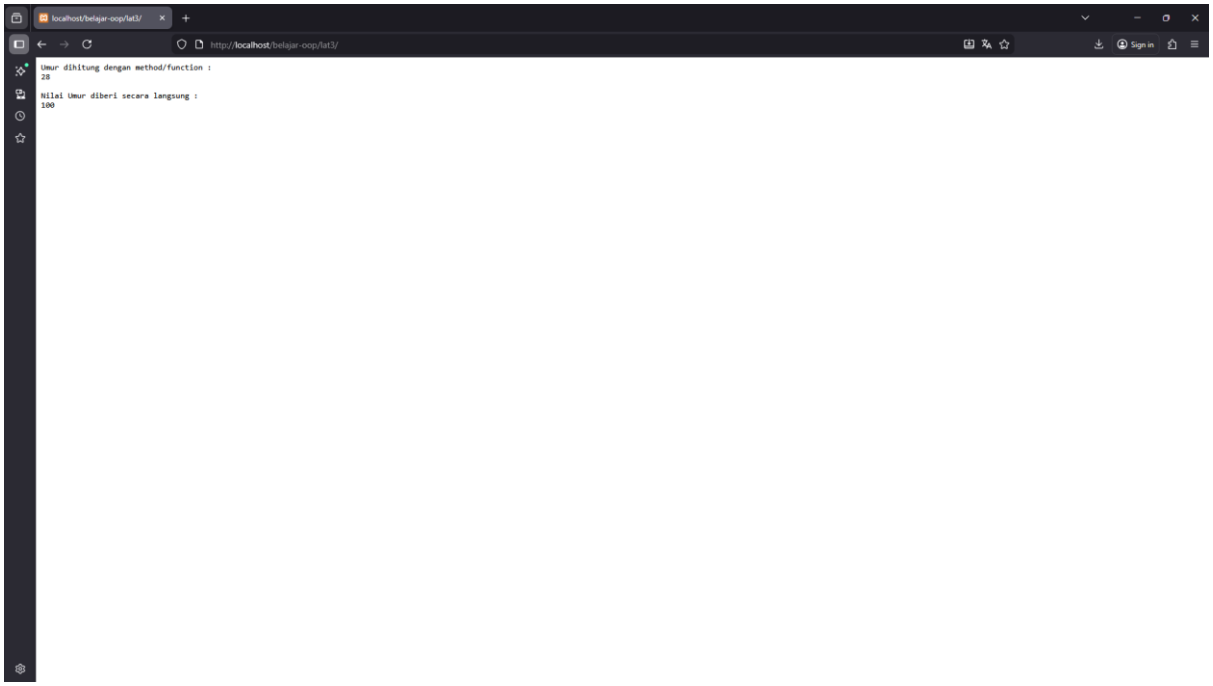
Code MataKuliah.php

```php
<?php
class MataKuliah
{
    private $kode;
    private $nama;

    public function setData($kode, $nama)
    {
        $this->kode = $kode;
        $this->nama = $nama;
    }

    public function getData()
    {
        return [$this->kode, $this->nama];
    }
}
```

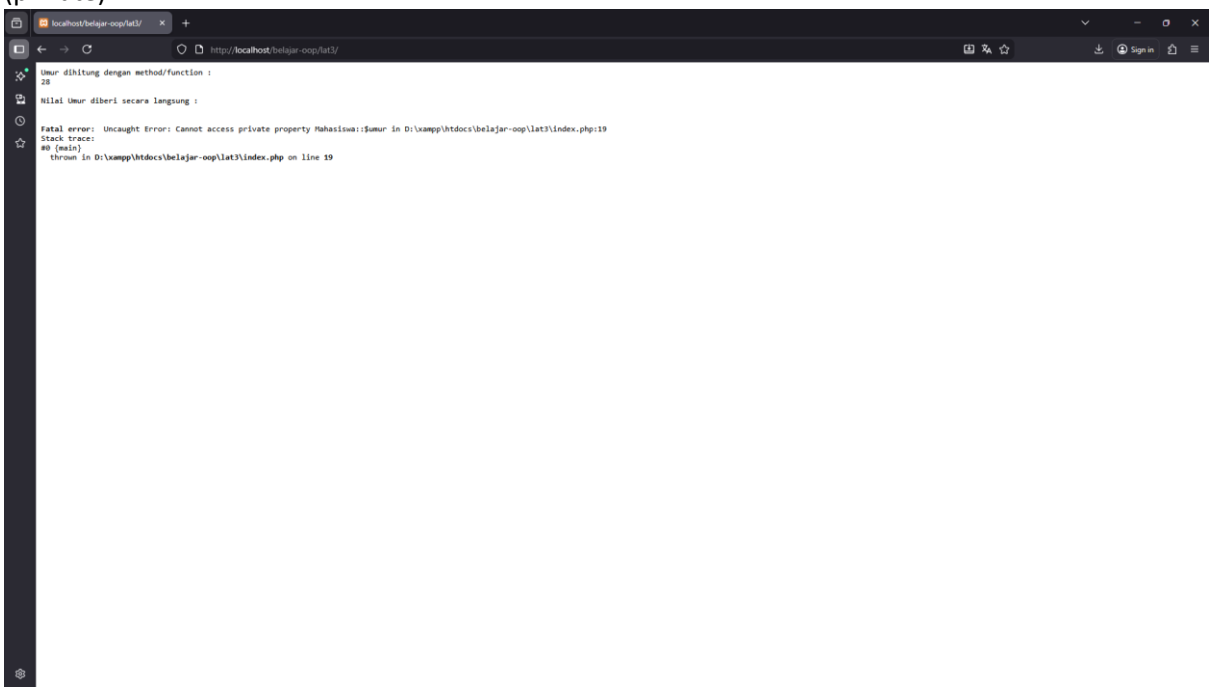
code index.php

```php
<?php
include "Mahasiswa.php";
include "MataKuliah.php";

$mahasiswa_1 = new Mahasiswa();
$makul_1 = new MataKuliah();
$makul_2 = new MataKuliah();

$makul_1->setData("A11.12345", "Pemrograman Berbasis Web");
$makul_2->setData("A11.12346", "Pemrograman Web Lanjut");

$mahasiswa_1->setData("A11.2023.12345", "Aprilyani Nur Safitri", [$makul_1, $makul_2], "1998-05-18");

echo "<pre>";
echo "Umur dihitung dengan method/function : <br>";
print_r($mahasiswa_1->getData()[4]);
echo "<br><br>";
echo "Nilai Umur diberi secara langsung : <br>";
$mahasiswa_1->umur = 100;
print_r($mahasiswa_1->umur);
echo "</pre>";
```

Hasilnya localhost/belajar-oop/lat3
(public)



(private)



Lat4

Code Mahasiswa.php

```php
<?php
class Mahasiswa
{
    private $nim, $nama, $matkul, $tgl_lahir, $status;
    private $umur;

    public function __construct()
    {
        $this->status = "Aktif";
    }

    public function setData($nim, $nama, $matkul, $tgl_lahir)
    {
        $this->nim = $nim;
        $this->nama = $nama;
        $this->matkul = $matkul;
        $this->tgl_lahir = $tgl_lahir;
        $this->setUmur();
    }

    public function setUmur()
    {
        $this->umur = date("Y") - substr($this->tgl_lahir, 0, 4);
    }

    public function getData()
    {
        return [
            "nim" => $this->nim,
            "nama" => $this->nama,
            "matkul" => $this->matkul,
            "tgl_lahir" => $this->tgl_lahir,
            "umur" => $this->umur,
            "status" => $this->status,
        ];
    }
}
```

$status protected

```php
<?php
class Mahasiswa
{
    private $nim, $nama, $matkul, $tgl_lahir;
    protected $status
    private $umur;

    public function __construct()
    {
        $this->status = "Aktif";
    }

    public function setData($nim, $nama, $matkul, $tgl_lahir)
    {
        $this->nim = $nim;
        $this->nama = $nama;
        $this->matkul = $matkul;
        $this->tgl_lahir = $tgl_lahir;
        $this->setUmur();
    }

    public function setUmur()
    {
        $this->umur = date("Y") - substr($this->tgl_lahir, 0, 4);
    }

    public function getData()
    {
        return [
            "nim" => $this->nim,
            "nama" => $this->nama,
            "matkul" => $this->matkul,
            "tgl_lahir" => $this->tgl_lahir,
            "umur" => $this->umur,
            "status" => $this->status,
        ];
    }
}
```

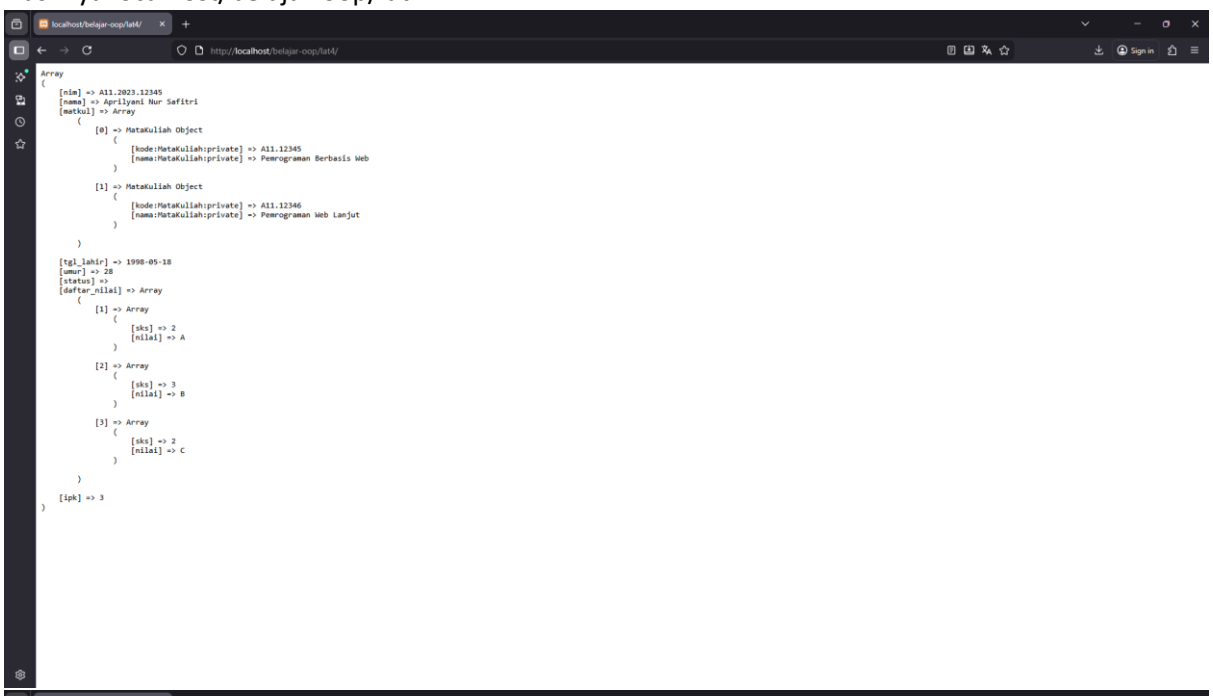
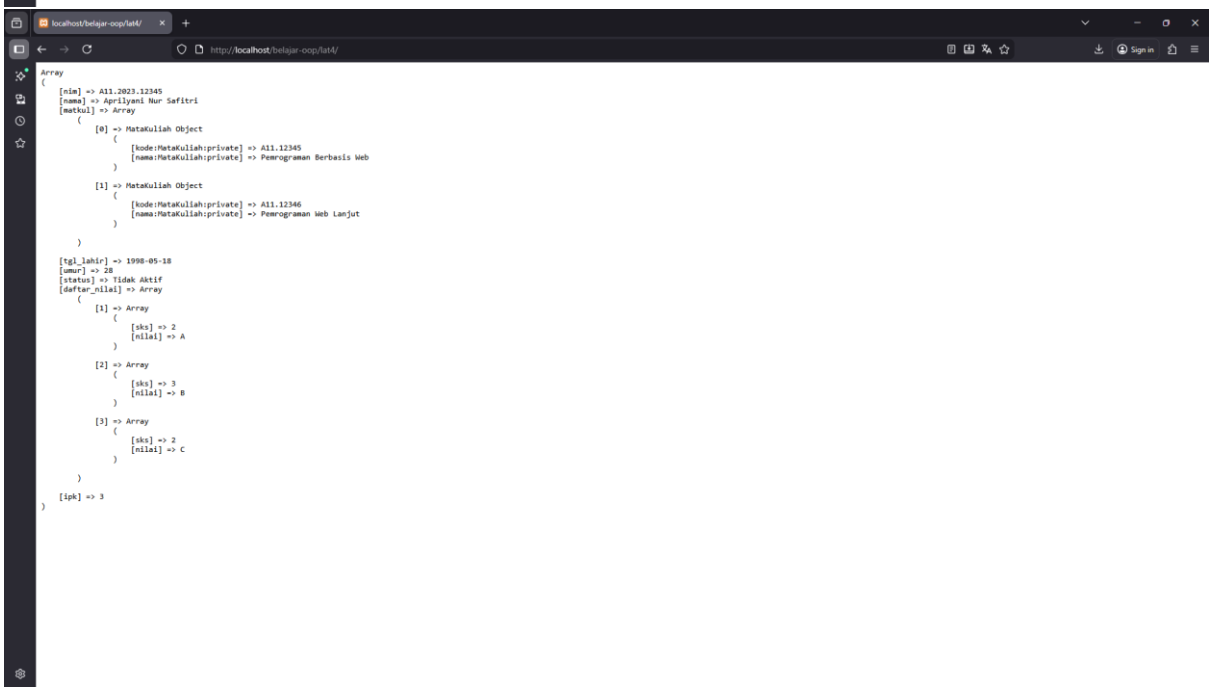
code MataKuliah.php

```php
<?php
class MataKuliah
{
    private $kode;
    private $nama;

    public function setData($kode, $nama)
    {
        $this->kode = $kode;
        $this->nama = $nama;
    }

    public function getData()
    {
        return [$this->kode, $this->nama];
    }
}
```

code index.php

```php
<?php
include "Mahasiswa.php";
include "MahasiswaTransfer.php";
include "MataKuliah.php";

$makul_1 = new MataKuliah();
$makul_2 = new MataKuliah();

$daftar_nilai = [
    "1" => ["sks" => 2, "nilai" => "A"],
    "2" => ["sks" => 3, "nilai" => "B"],
    "3" => ["sks" => 2, "nilai" => "C"],
];

$mahasiswa_1 = new MahasiswaTransfer($daftar_nilai);

$makul_1->setData("A11.12345", "Pemrograman Berbasis Web");
$makul_2->setData("A11.12346", "Pemrograman Web Lanjut");

$mahasiswa_1->setData("A11.2023.12345", "Aprilyani Nur Safitri", [$makul_1, $makul_2], "1998-05-18");

echo "<pre>";
print_r($mahasiswa_1->getData());
echo "</pre>";
```

code MahasiswaTransfer.php

```php
<?php
class MahasiswaTransfer extends Mahasiswa
{
    private $daftar_nilai, $ipk = 0, $ttl_sks = 0, $ttl_bobot = 0;
    private $bobot = ['A' => 4, 'B' => 3, 'C' => 2];

    public function __construct($daftar_nilai)
    {
        $this->daftar_nilai = $daftar_nilai;
        $this->getIPK();
        $this->status = 'Tidak Aktif';
    }

    public function getIPK()
    {
        if (!empty($this->daftar_nilai)) {
            foreach ($this->daftar_nilai as $data) {
                $this->ttl_sks += $data['sks'];
                $this->ttl_bobot += $data['sks'] * $this->bobot[$data['nilai']];
            }

            $this->ipk = $this->ttl_bobot / $this->ttl_sks;
        }

        return $this->ipk;
    }

    public function getData()
    {
        return array_merge(parent::getData(), [
            "daftar_nilai" => $this->daftar_nilai,
            "ipk" => $this->ipk,
        ]);
    }
}
```

Hasilnya localhost/belajar-oop/lat4